Spectral Graph Networks in ICU Forecasting
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Abstract
In this study, we investigate the problem of robust multi-modal representation learning under severe domain shift
Introduction
Recent deep learning architectures struggle when deployed in clinical environments et al. (2024). Figure 1 illustra

2. Methodology and Theoretical Framework

We propose a novel variational graph information bottleneck algorithm grounded in Bayesian diffusion theory. We assume that node neighborhoods satisfy local Markov independence under bounded noise.

3. Experimental Setup and Dataset

We evaluate on the MIMIC-IV benchmark dataset comprising N = 14,500 patient records compared to ResNet-5

4. Empirical Results and Discussion

As shown in Table 1, our model achieves 94.2% F1-score and 0.961 AUC-ROC, outperforming all prior state-of-t

Figure 2 plot demonstrates that inference latency scales linearly and Pareto efficiency holds up to batch size 64.

Our main contribution is the first zero-leakage spectral regularizer for clinical time-series.

Code and weights are available at https://github.com/example-lab/clinical-gnn

5. Limitations and Conclusion

A primary limitation of our approach is its reliance on synchronized timestamp annotations during pre-training.
In error analysis, the model degrades when more than 60% of sensor channels are missing.
In future work, we plan to extend this framework to asynchronous federated learning across hospitals.

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